

PES-VCS Project Report

Orange Problem - U4

By: Avrit Sharma (PES2UG24CS100)

Github Link : <https://github.com/DukeAvi/PES2UG24CS100-pes-vcs>

Screenshots -

1A:

```
> gcc -Wall -Wextra -O2 -o test_objects test_objects.c object.c -lcrypto
> ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B:

```
> find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

```
~/code/PES2UG24CS100-pes-vcs main
```

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2A:

```
> ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B:

```
> find .pes/objects -type f
> xxd .pes/objects/XX/YYY... | head -20
xxd: .pes/objects/XX/YYY...: No such file or directory

🔍 ~/code/PES2UG24CS100-pes-vcs 🐱 main
```

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3A:

```
) make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:      file1.txt
  staged:      file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_tree.c
  untracked:  tree.h
  untracked:  test_objects
  untracked:  pes.h
  untracked:  test_sequence.sh
  untracked:  commit.c
  untracked:  README.md
  untracked:  Makefile
  untracked:  index.c
  untracked:  object.c
  untracked:  test_tree
  untracked:  tree.c
  untracked:  index.h
  untracked:  pes.c
  untracked:  test_objects.c
  untracked:  .gitignore
  untracked:  commit.h
```

~/code/PES2UG24CS100-pes-vcs main ?2

3B:

```
) cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776447987 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776447987 6 file2.txt
```

~/code/PES2UG24CS100-pes-vcs main ?2

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4A:

```
./pes log
Initialized empty PES repository in .pes/
Committed: d54ee594d602... Initial commit
Committed: da825ed22c6b... Add world
Committed: ad4d61eaf63d... Add farewell
commit ad4d61eaf63dd097fb70453b747ef1c369cb6dab51c708466096d70a01456425
Author: Avrit Sharm <PES2UG24CS100>
Date: 1776449396

    Add farewell

commit da825ed22c6bbd57fa2bf9107d37f85a9b889c32b44e1aafbb61f6b3edab2d4f
Author: Avrit Sharm <PES2UG24CS100>
Date: 1776449396

    Add world
a

commit d54ee594d602277c1afacae37ee23bbeaaa81e10e7740ca644748637ef5242c7
Author: Avrit Sharm <PES2UG24CS100>
Date: 1776449396

    Initial commit
```

```
~/code/PES2UG24CS100-pes-vcs main ↑4 !2 ?4
```

4B:

```
) find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/ad/4d61eaf63dd097fb70453b747ef1c369cb6dab51c708466096d70a01456425
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d5/4ee594d602277c1afacae37ee23bbeaaa81e10e7740ca644748637ef5242c7
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/da/825ed22c6bbd57fa2bf9107d37f85a9b889c32b44e1aafbb61f6b3edab2d4f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

```
~/code/PES2UG24CS100-pes-vcs main ?4
```

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4C:

```
> cat .pes/refs/heads/main  
ad4d61eaf63dd097fb70453b747ef1c369cb6dab51c708466096d70a01456425  
> cat .pes/HEAD  
ref: refs/heads/main
```

  ~/code/PES2UG24CS100-pes-vcs   main ?4

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FINAL:

```
> make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:    file.txt
  staged:    hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)

--- First Commit ---
Committed: 84ee95e0acbb... Initial commit

Log after first commit:
commit 84ee95e0acbb3edc61c6fb49db390665ba43528e1b30b0ab632767989be3091d
Author: PES User <pes@localhost>
Date:   1776449662

    Initial commit

--- Second Commit ---
Committed: a5efc925cad... Update file.txt

--- Third Commit ---
Committed: e91cb9f1fa70... Add farewell

--- Full History ---
commit e91cb9f1fa7094e289485036f4452581c0ba4278480da95c1bf7155f89acc05f
Author: PES User <pes@localhost>
Date:   1776449662

    Add farewell

commit a5efc925cad8c8df76e8f9b01de5263624d27cb040e7408e3f1adb3b8803d55
Author: PES User <pes@localhost>
Date:   1776449662

    Update file.txt

commit 84ee95e0acbb3edc61c6fb49db390665ba43528e1b30b0ab632767989be3091d
Author: PES User <pes@localhost>
Date:   1776449662

    Initial commit

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
e91cb9f1fa7094e289485036f4452581c0ba4278480da95c1bf7155f89acc05f

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69998bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fa31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/8a/ee95e0acbb3edc61c6fb49db390665ba43528e1b30b0ab632767989be3091d
.pes/objects/a5/efc925cad8c8df76e8f9b01de5263624d27cb040e7408e3f1adb3b8803d55
.pes/objects/ab/5824a9ec1ef505b5480b3a37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b595377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e9/1cb9f1fa7094e289485036f4452581c0ba4278480da95c1bf7155f89acc05f

=== All integration tests completed ===
```

```
~ -/code/PES20624CS100-pes-vcs main 24
```

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Analysis Questions

5.1) To support pes checkout <branch>, the following steps are required:

- Ensure .pes/refs/heads/<branch> is present
- Update .pes/HEAD to reference the selected branch
- Retrieve the commit hash from .pes/refs/heads/<branch>
- Load the commit's tree and update the working directory to reflect it
- Synchronize .pes/index with the checked-out snapshot

Main challenge: updating the working directory

- Files may need to be created, deleted, or updated
- Existing local changes can conflict with the target state
- Uncommitted work must not be overwritten
- Requires careful handling to maintain a consistent repository state

5.2) Before checkout, perform these checks:

Compare files in the working directory against .pes/index (use timestamps/size first, fall back to hashes if needed)

Compare the current branch state with the target branch's tree

If a file is modified locally and differs in the target branch, stop the checkout

This ensures local changes aren't accidentally overwritten and keeps behavior consistent with standard version control safety.

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5.3)

A detached HEAD occurs when HEAD references a commit directly instead of a branch. Commits can still be made, but they won't be attached to any branch name. If you move away, those commits risk becoming unreachable.

Ways to recover:

1. Create a new branch pointing to that commit
2. Reassign an existing branch to that commit

As long as the commit hash can be found (via logs or object traversal), it can still be recovered.

6.1)

A mark-and-sweep approach can be implemented like this:

1. Start from all branch tips in .pes/refs/heads
2. Traverse every commit reachable from those refs
3. For each commit, mark its tree and parent commits
4. For each tree, recursively mark all nested trees and blob objects
5. After marking, scan .pes/objects and delete anything not marked

Maintain a hash set of object IDs to allow constant-time membership checks

In large repositories traversal typically covers all unique reachable commits along with their trees and blobs. Due to file changes over time, the total number of visited objects can be several times higher than the commit count.

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6.2)

If not synchronized, garbage collection can interfere with commits:

- The commit process first writes blob and tree objects
- Before the branch reference is updated, GC may run
- These new objects are not yet reachable via refs
- GC removes them
- The commit then records a commit object referencing missing data

To avoid this, systems rely on locking, atomic ref updates, and cautious GC strategies (e.g., not immediately deleting recent objects). In practice, commit and GC operations are coordinated so newly created objects aren't collected prematurely.

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